
OneVoice AI Challenge Updated Team Briefing

Everything we've learned since the architecture draft — and what it means for our plan

JUNE 13, 2026 · 11 DAYS TO REGISTRATION DEADLINE

The registration page revealed a different competition than we expected

What we thought

Open-ended innovation challenge — bring your best idea

Benchmarks unspecified, probably BLEU score only

Cloud dependency might be tolerated

Phone form factor is one option among many

What the Luma page actually says

STRICT BLEU + COMET accuracy, MOS by native speakers

HARD RTF < 1.0, turnaround latency < 2.0s with deductions

RED LINE Cloud dependency, disqualification

OPEN Any portable device — no hardware category restrictions

What changed most: the competition is a product-engineering hybrid with strict on-device requirements, not a science fair.

This is a product-engineering competition with academic evaluation rigor

The judges are looking for working prototypes with real measurements — not slides or theoretical models.

Qualcomm's DNA

Working on-device Snapdragon demos they can showcase. RTF < 1.0, sub-2s latency, no-cloud, field testing — all engineering gates.

Saigon AI Hub's DNA

Academically defensible winners. BLEU/COMET, MOS panel, proprietary benchmark dataset, published scoring criteria.

Dimension	Weight	What this means for us
Technical Excellence	50%	Measured latency, accuracy, stability on device — not estimates
Innovation & Novelty	25%	MeloTTS-VI Hub contribution > novel architecture
Business Viability	25%	Deployable on phones people already own — not custom hardware

Most competition winners stay at prototype stage — this contest is wired differently

The pattern

~9 out of 10 winners never commercialize

Winners pivot, join companies, or stall after media coverage

RTranslator (closest cousin): Whisper+NLLB on Android, 2k GitHub stars, never productized

Why this one is different

Market research confirms a real gap — not a synthetic problem

50% Technical weight forces engineering delivery, not slideware

Qualcomm + VNG have deployment incentives, not just PR

Finals require: working prototype + business case + pitch deck

The conclusion: a functional pipeline with real measurements puts us in the top tier.

Our edge is not invention — it's execution rigor on real hardware

The components are available to every team. What separates us is what we measure and engineer around them.

DeepFilterNet tonal hypothesis

Real WER numbers for Vietnamese under industrial noise. Fastest experiment, ownable data.

Thermal graceful degradation

Three planned fallback tiers. Most teams will overheat; we'll have a staged demo.

MeloTTS-VI Hub contribution

Shipping a Vietnamese TTS checkpoint back to Qualcomm's catalog. Institutionally aligned.

Real measurements vs estimates

Latency profiling, noise robustness, thermal calibration — all on SD8G2, not on paper.

"Specificity that can only come from having touched the problem" — §7.3 of the architecture doc

We are not building a luxury gadget — we are building a practical industrial tool

The wrong framing

"Near-native real-time translation"

Requires specialized \$300-500 hardware

Competing with Timekettle, iFLYTEK

Raises expectations we can't meet

The correct framing

"Good-enough translation on phones workers already own"

No specialized hardware — deploy via MDM

Shift handovers, safety briefings, skills transfer

This is the gap the contest problem statement identifies

Constraint-honest design: we're back-peddalling from the ideal to something that actually ships.

Phone-in-pocket beats wearables for sustained Edge AI

The "hands-free" concern

Contest says "hands-busy" environments — not a wearable requirement

No rule mandates hands-free device format

Phone on lanyard + VAD trigger + Bluetooth earpiece = hands remain free

Phone vs. wearable compute

Wearables: tiny battery, weak NPU, overheat under continuous AI load

Phone (SD8G2): 16GB RAM, dedicated NPU, thermal mass for hours of inference

Zero additional hardware cost in enterprise deployment

50M+

SD8G2 devices shipped (2023 flagships)

Thermal sweet spot: TSMC 4nm, mature, stable

\$2-9

Per-device/month MDM cost in enterprise

New discoveries change the component plan — mostly to our advantage

Layer	Old plan	New finding	Impact
TTS	Train MeloTTS-VI from scratch	Community checkpoint exists (MIT, ~25h data, PhoBERT-based)	Saves months — use as base, fine-tune further
MT	NLLB-600M int8	Hy-MT1.5 from Tencent (440MB, 33 languages, beats 72B models)	Backlog. Needs SD8G2 latency testing
MT License	CC-BY-NC-4.0 (flag raised)	Confirmed: non-commercial license conflicts with \$20K prize	Needs resolution before Phase 2
ASR	Whisper Medium via Hub	Whisper-Medium reinstated v0.51, Voice AI runtime added	Better export path, no change needed

MeloTTS-VI checkpoint is the biggest win: we don't start from zero.

TIMELINE

We have 11 days to registration — target June 18 with buffer to June 24

PHASE 1 · DAY 1 (JUN 13)

Decisions: solo/team, project name

GitHub repo + Google Sheets template

PHASE 2 · DAY 2-3 (JUN 14-15)

Pitch deck draft

DeepFilterNet WER experiment running

PHASE 3 · DAY 4-5 (JUN 16-17)

Code scaffolding pushed to GitHub

Luma long answers drafted

Basic ASR, MT, TTS chain on laptop

SUBMIT · DAY 6 (JUN 18)

Submit early — free Phase 2 focus

Buffer Jun 19-24 if not ready

The core thesis

We are the first team to build a complete, field-tested, Snapdragon-NPU-validated speech-to-speech translation pipeline with Vietnamese as a primary language, using exclusively open models that extend the Qualcomm AI Hub catalog.

Three decisions to make today — then we build

Today's decisions

1. Solo or team?

If team: who, what roles, who submits the form

2. Project name?

Memorable, available on GitHub, pitch-worthy

3. Pitch deck style?

Technical or investor-facing? Sets tone for registration

Let's ship.

This week's focus

DeepFilterNet experiment — highest evidence-to-effort ratio. CPU-runable. WER comparison table possible by Wednesday.

Pipeline scaffolding — Silero VAD + Whisper + NLLB/MeloTTS stubs on GitHub. Shows architecture alive.

Luma answers draft — 7 long-text fields. Source material in architecture.md.